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**DUNMAN HIGH SCHOOL**  
**Preliminary Examination**  
**Year 6**

**H2 BIOLOGY**

Paper 2 Structured Questions

9744/02

**14 September 2017**

**2 hours**

**INSTRUCTIONS TO CANDIDATES:**

DO NOT TURN THIS PAGE OVER UNTIL YOU ARE TOLD TO DO SO.

READ THESE NOTES CAREFULLY.

Answer **all** questions.

Write your answers on space provided in the Question Paper.

**INFORMATION FOR CANDIDATES**

Essential working must be shown.

The intended marks for questions or parts of questions are given in brackets [ ].

| For Examiner's Use |             |
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| <b>1</b>           | <b>/ 8</b>  |
| <b>2</b>           | <b>/10</b>  |
| <b>3</b>           | <b>/9</b>   |
| <b>4</b>           | <b>/ 11</b> |
| <b>5</b>           | <b>/ 10</b> |
| <b>6</b>           | <b>/ 11</b> |
| <b>7</b>           | <b>/ 8</b>  |
| <b>8</b>           | <b>/ 12</b> |
| <b>9</b>           | <b>/ 4</b>  |
| <b>10</b>          | <b>/ 7</b>  |
| <b>11</b>          | <b>/ 10</b> |
| <b>Total [100]</b> |             |

This document consists of **32** printed pages and **4** blank pages.

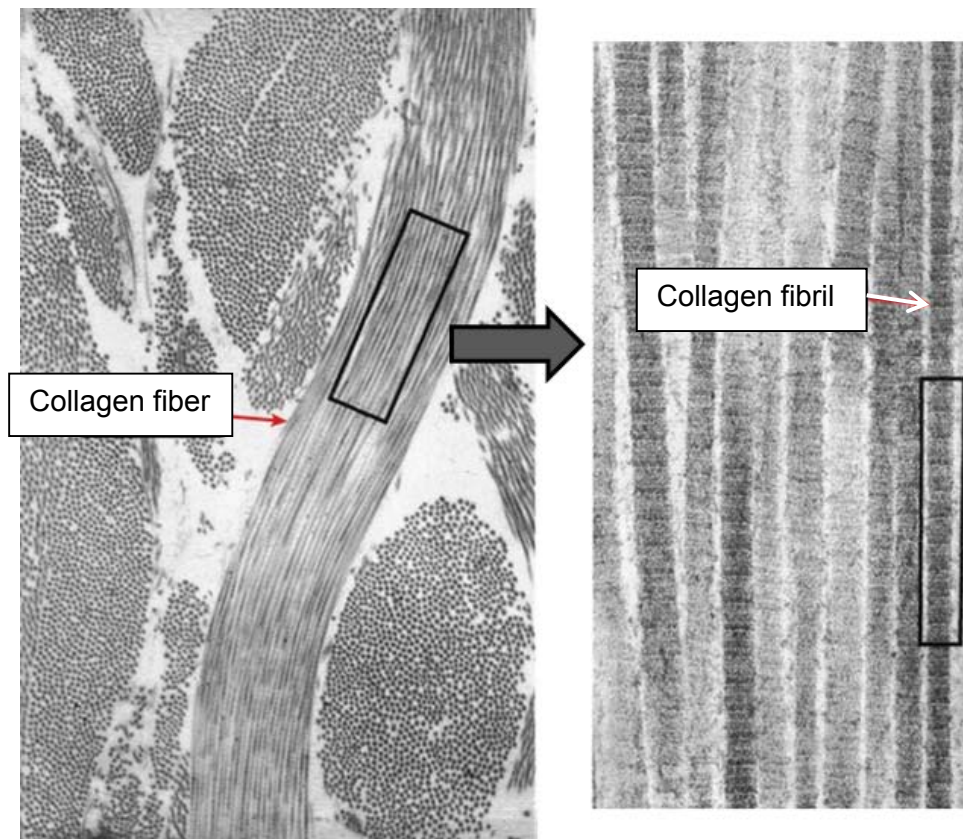
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**Structured Questions (100 marks)**  
Answer **all** questions in this section.

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**Question 1**

Collagen is the main structural protein of the various connective tissues in animals. As the main component of connective tissue, it is the most abundant protein in mammal, making up from 25% to 35% of the whole-body protein content. **Fig. 1.1** shows the structure of a collagen fiber and collagen fibrils.



**Fig. 1.1**

- (a)** Describe the primary structure of a collagen polypeptide. [2]

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**(b)** With reference to **Fig. 1.1**,

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- (i)** Describe two ways how the structure of a collagen fiber differs from that of a DNA molecule. [2]

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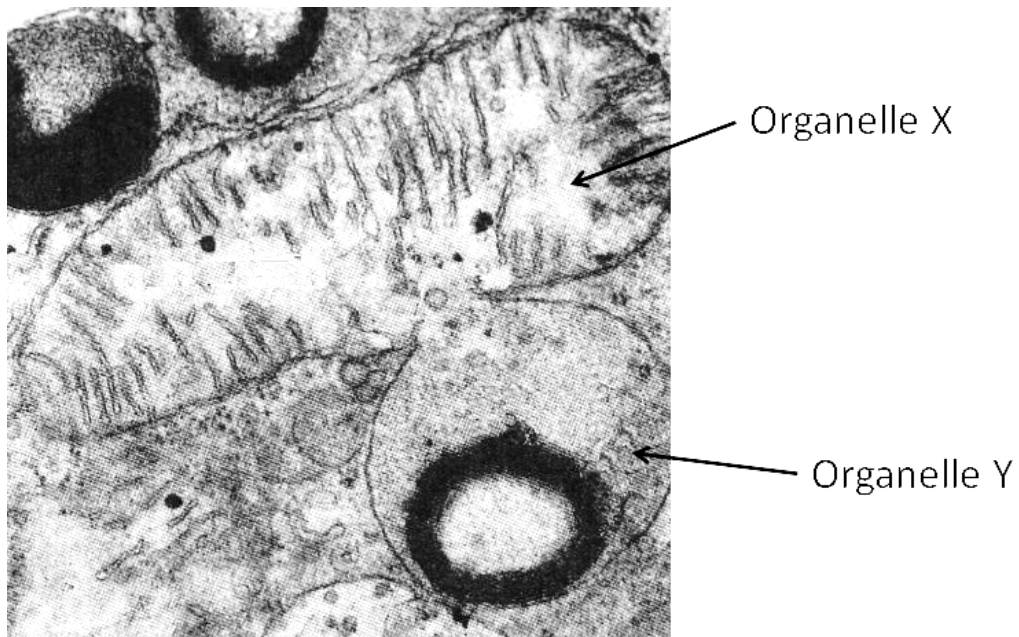
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- (ii)** Draw a diagram to show why a collagen fibril has high tensile strength. Annotate your diagram appropriately. [2]

(c) Fig. 1.2 shows two membrane-bound organelles commonly found in animal cells.

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**Fig 1.2**

Describe two other functions of organelle Y not shown in **Fig. 1.2**. [2]

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**Total: [8]**

**Question 2***For  
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A research group discovers a hydrolytic enzyme in the aleurone of the barley seed that converts lipids to simpler molecules and begins to characterize it. Aleurone is a protein found in protein granules of maturing seeds and tubers.

The researchers found that the three essential catalytic groups in the active site are contributed by histidine 57, aspartate 102 and serine 195, where the numbers denote the position of the amino acid in the amino acid sequence.

- (a) (i)** Describe one advantage of storing lipids rather than starch in a seed. [1]

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- (ii)** Explain what determines the precise position of these three amino acids in the structure of the hydrolytic enzyme. [3]

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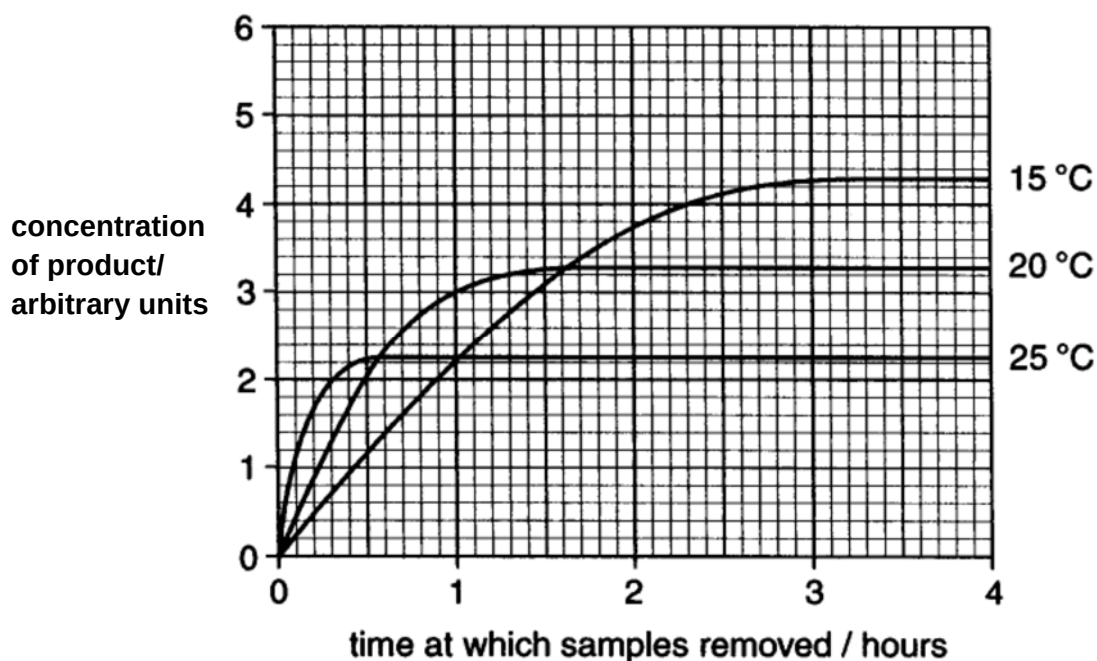
The researchers discovered that the seed germinates best if the soil is cool at close to

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5°C. In an investigation into the properties of the hydrolytic enzyme, they set up three water baths at 15, 20 and 25°C. Into each bath was placed a tube containing 1 cm<sup>3</sup> of the enzyme solution and a tube containing 10 cm<sup>3</sup> of concentrate substrate solution. On reaching the required temperature, the enzyme and substrate were quickly mixed and kept in the water bath.

There was a large excess of the substrate, so that the substrate concentration was not a limiting factor.

Samples were taken from each tube at regular intervals and the concentration of the product in these samples was determined. The results are shown in **Fig. 2**.



**Fig. 2**

**(b) (i)** Account for the curve at 15°C in **Fig. 2**. [3]

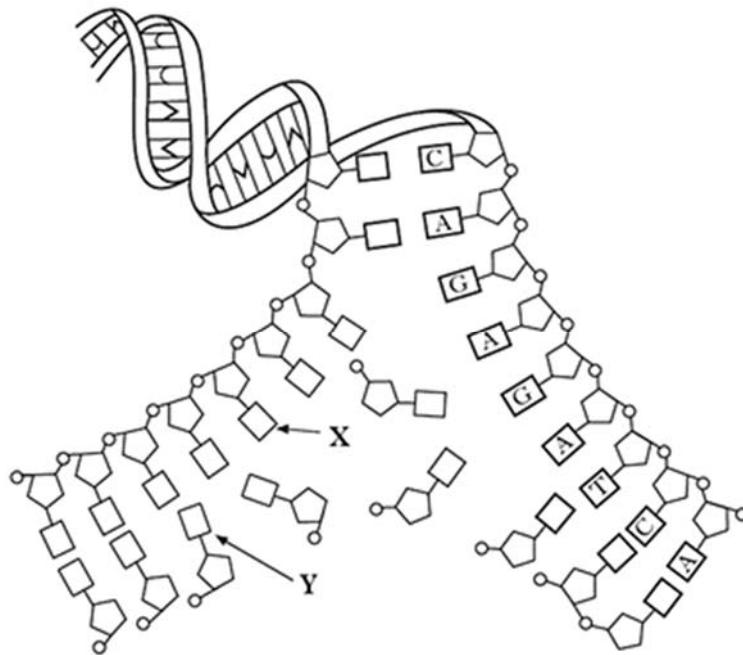
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- (ii) Predict and explain the effect of carrying out the procedure at 5°C. Sketch your prediction on **Fig. 2**. Label it as **Y**. [3]

**Total:[10]**

(a) **Fig. 3** shows replication of a part of the glucagon receptor gene.



**Fig. 3**

(i) Name the bases labelled X and Y on **Fig. 3**. [1]

X

Y

(ii) Explain how **Fig. 3** shows semi-conservative replication DNA. [3]

(b) DNA replication involves a number of enzymes including DNA polymerase.



Describe two other enzymes and their functions in DNA replication. [2]

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**(c)** Contrast the elongation stage in DNA replication with translation. [3]

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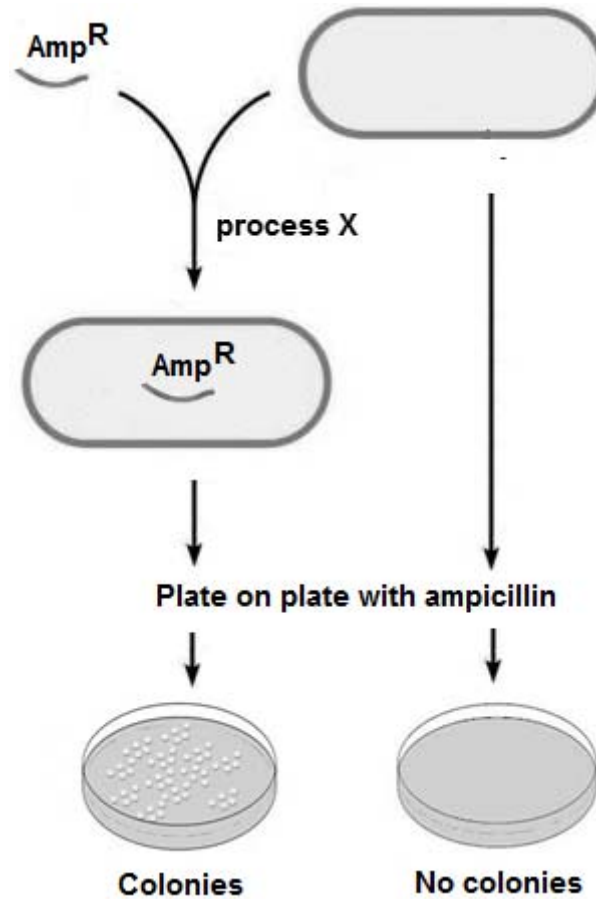
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**Question 4**

For

A student wanted to introduce ampicillin resistance gene ( $\text{Amp}^R$ ) to a strain of bacteria. **Fig. 4.1** shows a drawing done by the student to summarize his experimental procedure and expected results.



**Fig. 4.1**

- (a) (i) What is process X? [1]

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- (ii) What is the significance of the plate with no colonies? [2]

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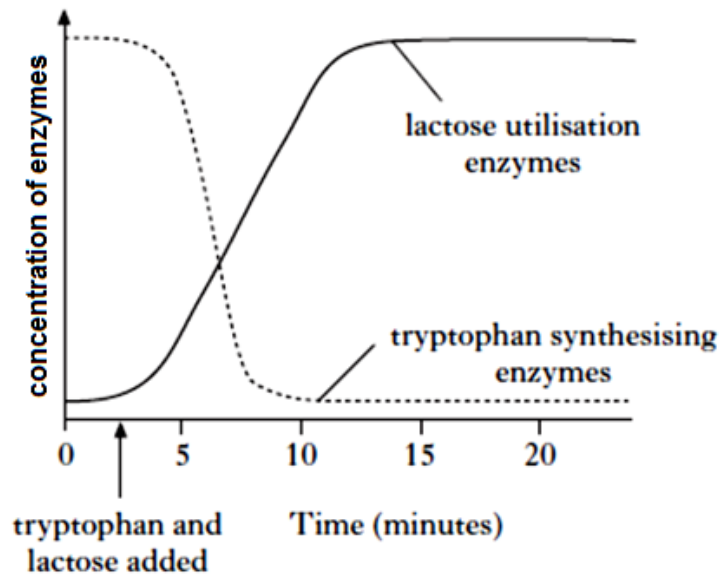
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- (iii) When the student carried out the experiment, he did not obtain any colonies on either plates. Explain why. [1]

In bacteria, genes coding for enzymes involved in the same metabolic pathway are arranged into operons. **Fig. 4.2** shows the changes in the concentration of enzymes that synthesise tryptophan and utilise lactose in a bacteria cell after the addition of tryptophan and lactose.



**Fig. 4.2**

- (b) (i) Describe the difference in the shape of the graph after the introduction of tryptophan and lactose. [2]

- (ii) Explain the change in concentration of tryptophan synthesizing enzyme after the introduction of tryptophan. [3]

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- (c) Name one lactose utilisation enzyme and suggest why the bacteria cells maintain some of this enzyme before the introduction of lactose? [2]

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## Question 5

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Fig. 5 shows the early development of a human embryo after fertilisation.

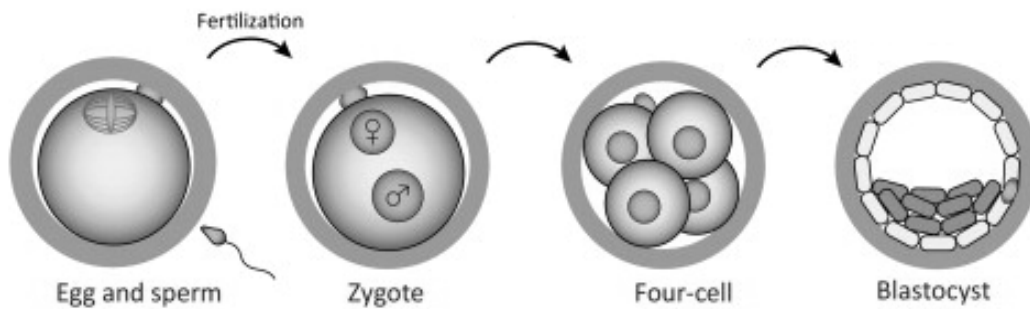
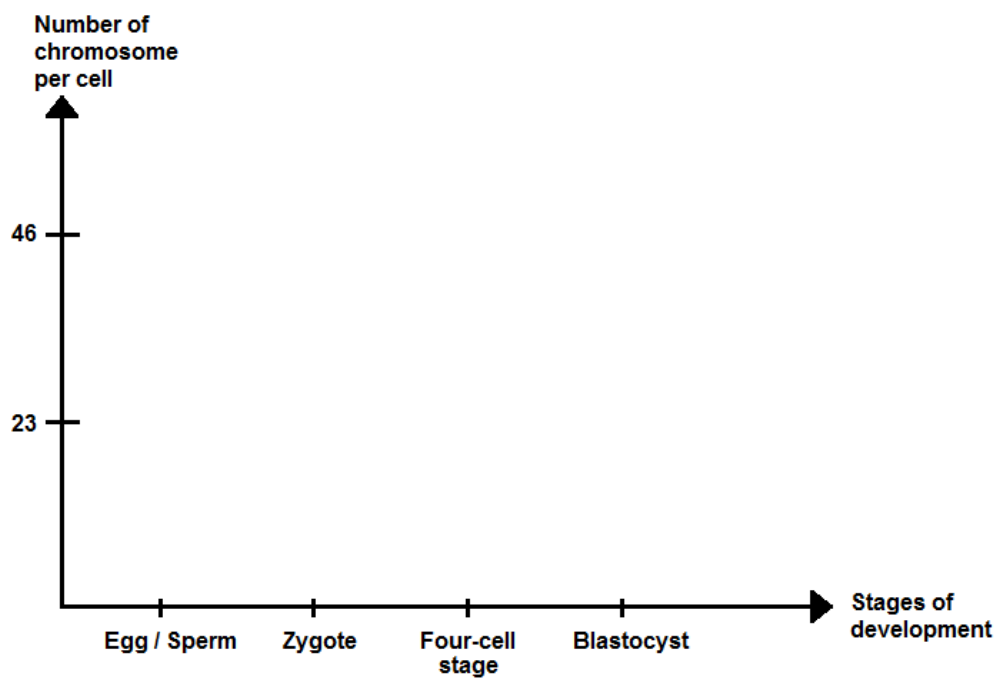


Fig. 5

- (b) (i) Name the type of cell division undergone by the zygote to form the four-cell stage. [1]

- (ii) Plot accurately, in the graph below, the number of chromosome per cell for the four stages of development and complete with a line graph. [2]



**(b) (i)** Explain the term “*asymmetrically*”. [1]

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### Table 5.1

|                     |                       |
|---------------------|-----------------------|
| Hb A $\beta$ globin | thr – pro – glu ..... |
| Hb S $\beta$ globin | thr – pro – val ..... |

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Table 5.2 shows the DNA triplet code.

Table 5.2

|              |   | Second Letter                            |                                      |                                            |                                           |                  |
|--------------|---|------------------------------------------|--------------------------------------|--------------------------------------------|-------------------------------------------|------------------|
|              |   | T                                        | C                                    | A                                          | G                                         |                  |
| First Letter | T | TTT } Phe<br>TTC }<br>TTA } Leu<br>TTG } | TCT }<br>TCC } Ser<br>TCA }<br>TCG } | TAT } Tyr<br>TAC }<br>TAA Stop<br>TAG Stop | TGT } Cys<br>TGC }<br>TGA Stop<br>TGG Trp | T<br>C<br>A<br>G |
|              | C | CTT }<br>CTC } Leu<br>CTA }<br>CTG }     | CCT }<br>CCC } Pro<br>CCA }<br>CCG } | CAT } His<br>CAC }<br>CAA Gln<br>CAG }     | CGT }<br>CGC } Arg<br>CGA }<br>CGG }      | T<br>C<br>A<br>G |
|              | A | ATT }<br>ATC } Ile<br>ATA }<br>ATG Met   | ACT }<br>ACC } Thr<br>ACA }<br>ACG } | AAT } Asn<br>AAC }<br>AAA Lys<br>AAG }     | AGT } Ser<br>AGC }<br>AGA } Arg<br>AGG }  | T<br>C<br>A<br>G |
|              | G | GTT }<br>GTC } Val<br>GTA }<br>GTG }     | GCT }<br>GCC } Ala<br>GCA }<br>GCG } | GAT } Asp<br>GAC }<br>GAA Glu<br>GAG }     | GGT }<br>GGC } Gly<br>GGA }<br>GGG }      | T<br>C<br>A<br>G |

With reference to Table 5.1 and 5.2,

- (ii) explain the **minimum** number of mutation that resulted in HbS. [2]

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- (iii) Identify the **maximum** number of gene mutations HbA can undergo such that the polypeptide sequence shown in Fig 5.1 is unchanged. [1]

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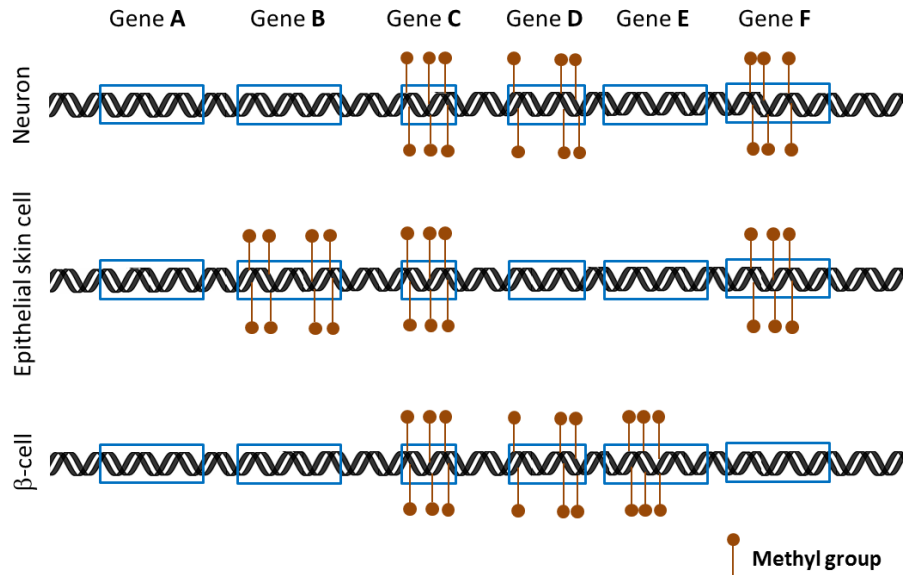
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## Question 6

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- (a) **Fig. 6.1** shows the methylation pattern of a segment of DNA from a human chromosome taken from three different cell types.



**Fig. 6.1**

- (i) With reference to the **Fig. 6.1**, describe and explain the effect of DNA methylation on gene expression in the neuron. [4]

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- (ii) Gene **D** codes for melanin which is a type of pigment found in most organisms. Melanin is produced only in skin cells and functions as the primary determinant of skin colour.

Explain why the pattern of DNA methylation in gene **D** differs in the three cell types. [3]

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- (iii) Suggest why gene **A** is not methylated in all the cell types and hence suggest a possible identity of gene **A**. [2]

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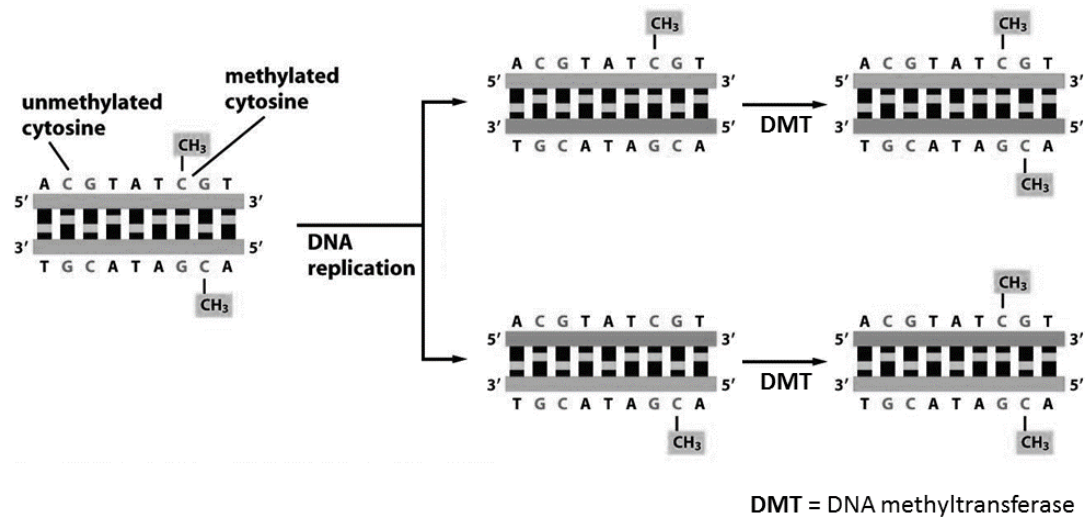
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- (b) Methylation of DNA always occurs on the cytosine base of a CG dinucleotide. The methylation pattern is heritable when a cell divides by mitosis. **Fig. 6.2** illustrates this process.

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**Fig. 6.2**

Using information in **Fig. 6.2**, suggest how it is possible for the methylation pattern to be inherited. [2]

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**Total:[11]**

**Question 7**

Cats possess a gene for producing tails. The tailless Manx phenotype in cats is produced by an allele that is lethal in the homozygous state. The Manx allele  $M^L$  severely interferes with normal spinal development. In heterozygotes ( $M^L M$ ), this results in the absence of tail.

Female cats are homogametic while male cats are heterogametic. The gene for black/orange/tortoiseshell coat colour is located on X chromosome and has two alleles  $X^O$  and  $X^o$ . Table below shows the genotypes of cats of different colours.

|                  |                                                                     |
|------------------|---------------------------------------------------------------------|
| $X^o X^o, X^o Y$ | Black coated female, male                                           |
| $X^O X^O, X^O Y$ | Orange coated female, male                                          |
| $X^O X^o$        | Tortoiseshell (intermingled black and orange in fur) in female only |

The table below shows the genotypes of two cats.

|                    | Female cat      | Male cat      |
|--------------------|-----------------|---------------|
| <b>Coat colour</b> | Orange          | black         |
| <b>tail</b>        | No tail         | No tail       |
| <b>Genotype</b>    | $X^O X^O M^L M$ | $X^o Y M^L M$ |

- (a) List down all possible genotypes of their zygotes if these two cats were to mate. [2]

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48 offspring were obtained from the above cross and the phenotype of the offspring were as follow:

| Number | Phenotype                         |
|--------|-----------------------------------|
| 17     | Tortoiseshell female with no tail |
| 7      | Tortoiseshell female with tail    |
| 14     | Orange male with no tail          |
| 10     | Orange male with tail             |

John and Mary were studying the genetic inheritance of Manx and coat colour in cats and each of them came to a different conclusion about the expected ratio of phenotypes of offspring from the cross of these two cats. Below are their respective conclusions.

#### John's conclusion

| Ratio | Phenotype                         |
|-------|-----------------------------------|
| 3     | Tortoiseshell female with no tail |
| 1     | Tortoiseshell female with tail    |
| 3     | Orange male with no tail          |
| 1     | Orange male with tail             |

#### Mary's conclusion

| Ratio | Phenotype                         |
|-------|-----------------------------------|
| 2     | Tortoiseshell female with no tail |
| 1     | Tortoiseshell female with tail    |
| 2     | Orange male with no tail          |
| 1     | Orange male with tail             |

| Degree of freedom | Probability, $p$ |      |       |       |       |
|-------------------|------------------|------|-------|-------|-------|
|                   | 0.1              | 0.05 | 0.02  | 0.01  | 0.001 |
| 1                 | 2.71             | 3.84 | 5.41  | 6.64  | 10.83 |
| 2                 | 4.61             | 5.99 | 9.21  | 9.21  | 13.82 |
| 3                 | 6.25             | 7.82 | 11.35 | 11.35 | 16.27 |
| 4                 | 7.78             | 9.49 | 13.28 | 13.28 | 18.47 |

$$\chi^2 = \sum \frac{(O - E)^2}{E} \quad v = c - 1$$

where  $\Sigma$  = 'sum of...'

O = observed 'value'

$v$  = degrees of freedom

E = expected 'value'

$c$  = number of classes

Using the formula for the  $\chi^2$  test and the table of  $\chi^2$  values above,

- (b) (i) Calculate the value of the  $\chi^2$  test based on John's conclusion. Show your working in the space provided below. [2]
- (ii) Calculate the value of the  $\chi^2$  test based on Mary's conclusion. Show your working in the space provided below. [2]

(c) In your opinion, who do you think is **MORE CORRECT**? Explain your choice. [2]

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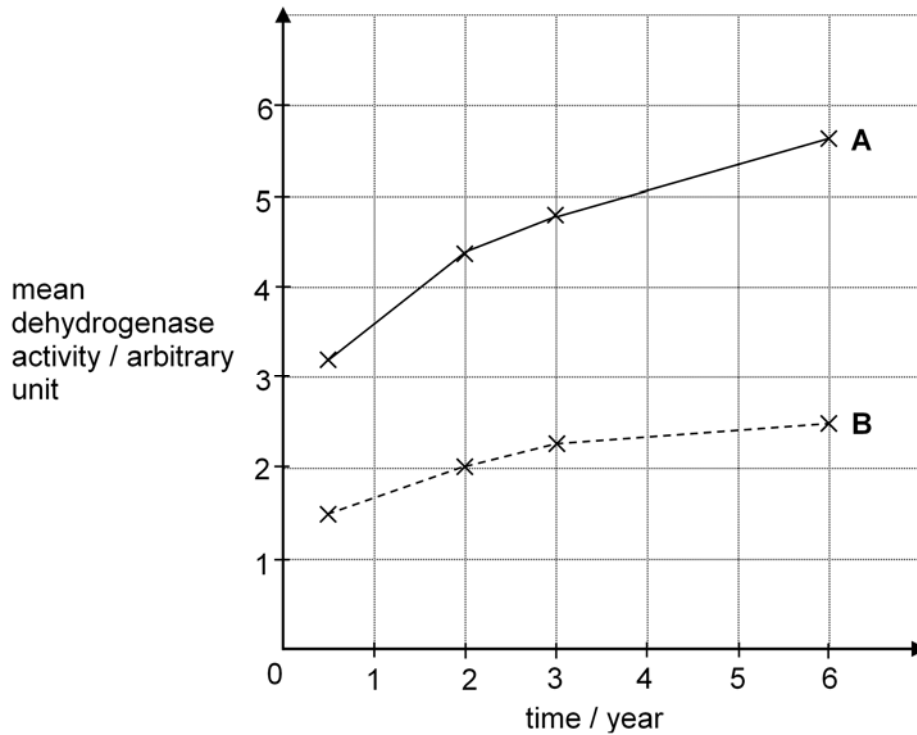
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**Question 8**

Studies were carried out on soil-dwelling aerobic bacteria. Soil samples were taken at two depths, **A** and **B**. The samples were taken at intervals over six years to determine the activity of dehydrogenases, involved in the Krebs cycle.

**Fig. 8** shows the mean dehydrogenase activity of the bacteria in these samples.



**Fig. 8**

- (a) (i) Explain the importance of Krebs cycle dehydrogenase in ATP synthesis. [3]

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- (ii) With reference to **Fig. 8** and your knowledge on enzymes, explain which samples, **A** or **B**, were taken from a greater depth. [4]

*For  
Examiner's  
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- (b) Dehydrogenase is also required for anaerobic respiration. Describe the process catalysed by the lactate dehydrogenase. [2]

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- (c)** Photosynthetic bacteria can be found in the ocean. Samples of bacteria were collected at the same depth from different locations and the activity of the enzyme RUBISCO was studied. Results obtained show that the samples collected near factories had higher RUBISCO activities than samples collected near forests.

- (i)** Identify the factor which explains the differing result. [1]

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- (ii)** Explain how the factor mentioned in **(c)(i)** affects the activity of RUBISCO in samples near factories. [2]

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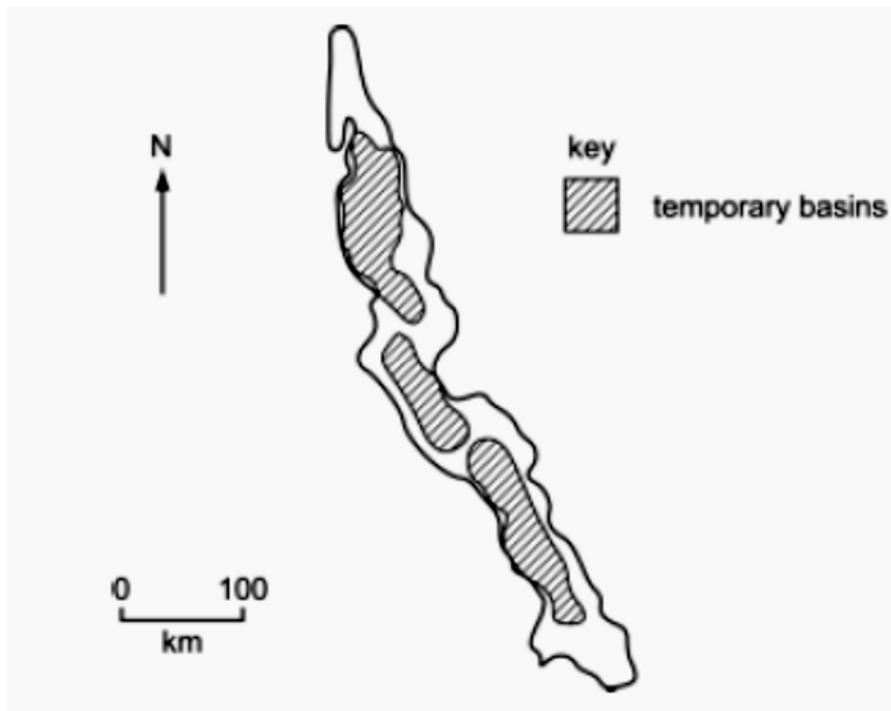
**Question 9**

In Lake Tanganyika in Africa, there are six species of fish of the genus *Tropheus* and a much larger number of distinctly coloured subspecies of each of the six species. *Tropheus* species are small fish that are confined to isolated rocky habitats around the shores of Lake Tanganyika.

The six species evolved during the primary radiation phase when the lake was first filled, about 1.25 million years ago. They arose from river dwelling ancestors and then filled all available niches in the lake.

Secondary radiations into the many subspecies occurred during the last 200 000 years. Sometime during this period, the water level in the lake fell, resulting in the formation of three separate lake basins. These basins persisted for many thousands of years before the water level rose again.

**Fig. 9** shows an outline map of the lake and the location of the three temporary basins caused by lowering of lake levels.



**Fig. 9**

Explain how natural selection could have caused the evolution of the six closely related species in the primary radiation. [4]

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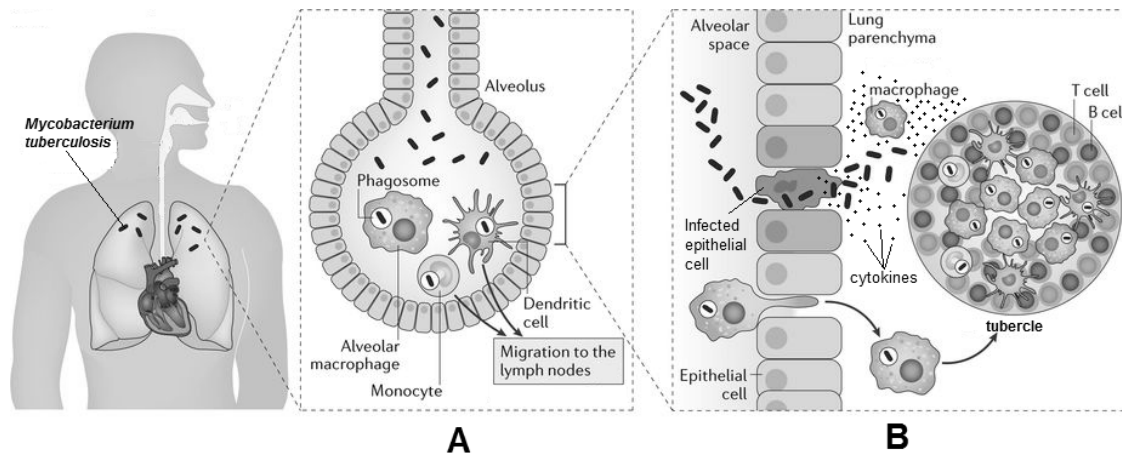
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**Total:[4]**

## Question 10

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**Fig. 10** shows the infection of *Mycobacterium tuberculosis* in latent Tuberculosis infection (LTBI).



**Fig. 10**

(a) Describe how *M. tuberculosis* is transmitted. [2]

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(b) (i) With reference to **Fig. 10A**, explain why the monocytes and dendritic cells migrate to the lymph nodes. [2]

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- (ii) With reference to **Fig. 10B** and using your own knowledge, describe how the tubercle is formed. [3]

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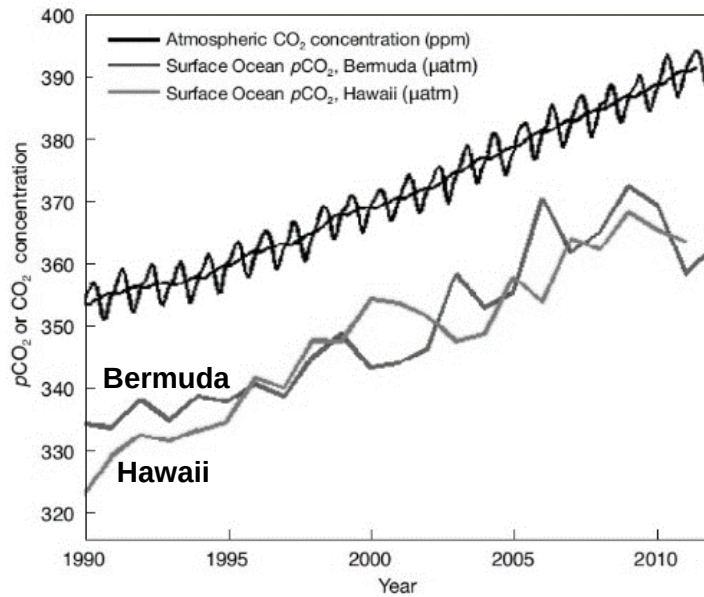


**Question 11**

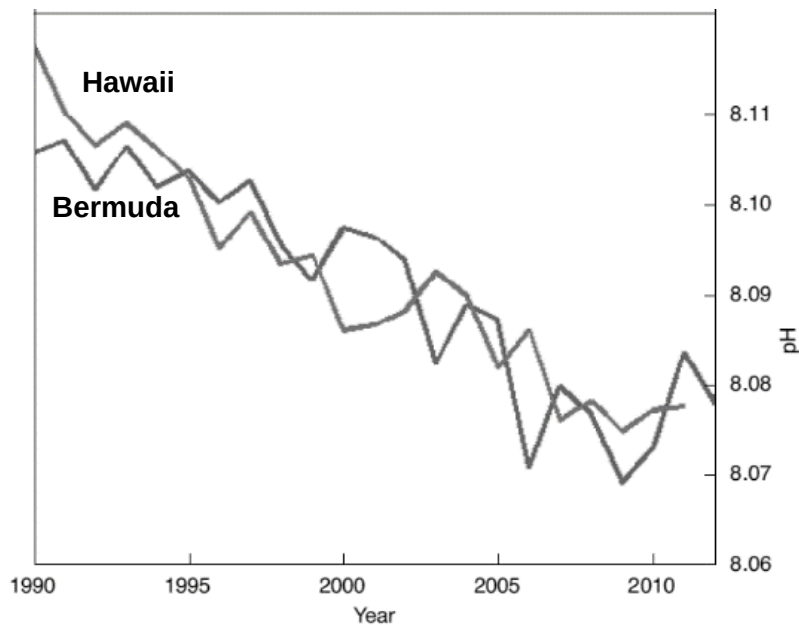
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Surface ocean carbon dioxide concentration can be determined by recording the concentration of carbon dioxide, in a closed volume of air that was circulated with a constantly renewed supply of water obtained two to three meters below the surface of the ocean.

**Fig. 11.1A** and **B** are graphs showing the changes in concentration of carbon dioxide in the air and changes in pH in the oceans of Bermuda and Hawaii from 1990 to 2010.



**Fig 11.1A**



**Fig 11.1B**

- (a) (i) Explain three human activities that have resulted in increased emission of carbon dioxide into the atmosphere. [3]

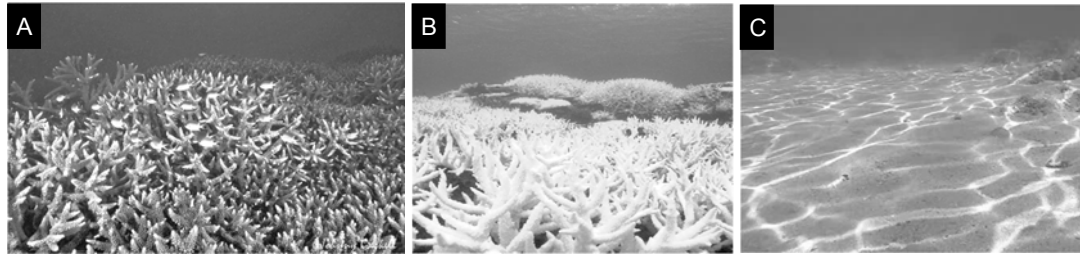
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- (ii) With reference to **Fig. 11.1A** and **B**, describe and explain the relationship between atmospheric CO<sub>2</sub> concentration and ocean pH for both Hawaii and Bermuda. [3]

- (b) The increase in carbon dioxide in the atmosphere causes warming of the Earth. The ocean absorbs most of these excess heat from the atmosphere. The top few meters of the ocean stores as much heat as Earth's entire atmosphere.

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**Fig. 11.2** shows examples of reefs from the Great Barrier Reef in different concentrations of surface ocean carbon dioxide.



|                                                     | A   | B       | C    |
|-----------------------------------------------------|-----|---------|------|
| Temperature increase / °C                           | +1  | +2      | +3   |
| Concentration of surface ocean carbon dioxide / ppm | 375 | 450-500 | >500 |

**Fig. 11.2**

With reference to **Fig 11.2**, describe and explain the effect of increasing concentrations of surface ocean carbon dioxide on coral reefs. [4]

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**DUNMAN HIGH SCHOOL  
PRELIMINARY EXAMINATION 2017  
YEAR SIX  
H2 BIOLOGY (9744)**

**Suggested Answers**

**Question 1**

**(a)**

- Primary structure made up of amino acids joined together by peptide bonds
- Contains proline and glycine, hydroxyproline and hydroxylysine
- very third amino acid in the polypeptide sequence is glycine

**(b)(i)**

- A collagen fiber is made up of amino acids while DNA molecule is made up of deoxyribonucleotides
- A collagen fiber is made up of the aggregation of many polypeptide chains cross-linked together while DNA molecule is made up of 2 strands of DNA

**(b)(ii)**

- Staggered arrangement of tropocollagen, label tropocollagen
- Label covalent cross-linkages between tropocollagen

**(c)**

- fuse with endocytotic vesicles/ food vacuole to digest the materials within
- Release hydrolytic enzymes resulting in self-digestion of the cell / autolysis

**Question 2**

**(a)(i)**

- For the same weight/mass, lipid has a higher / twice the amount of energy than starch thus allowing less amount of fats to be stored for the same amount of energy

**(a)(ii)**

- due to a specific amino acid sequence which result in precise/specific folding into secondary and tertiary structure
- Bringing R groups of amino acids far apart to close proximity in the active site
- Interactions between R groups results in formation of hydrogen, ionic and disulphide bonds and hydrophobic interactions

**(b)(i)**

- Linear increase of concentration of product from 0 to 3.8 a.u as time increase from 0 to 2hr
- enzyme is still working at 15 °C even though optimum temperature is 5 °C
- beyond 2hr, curve starts to level off and plateau at 3 hr and concentration of product remain at 4.3 a.u
- at this point, the enzyme is denatured, because at 15 °C, heat energy causes the ionic and hydrogen bonds to be broken

**(b)(ii)**

Explanation: [2]

- Rate of reaction is slower because lower kinetic energy compared to the other temperatures. Lower number of effective collisions between enzyme and substrate, resulting in less enzyme-substrate complexes formed and hence less product formed
  - Concentration of product is higher than for the other temperatures because it is the optimum temperature at which the hydrolytic enzyme works
  - Graph: [1]
- Rate of reaction slower than 15°C, should not show plateau

**Question 3****(a)(i)**

X – Cytosine

Y – Thymine

**(a)(ii)**

- parental strand acts as template for the synthesis of the new strand
- parental strand CAGAGATCA will result in the newly synthesised strand with sequences GTCTCTAGT
- newly synthesised daughter DNA molecule consists of one original strand and one newly synthesised strand

**(b)**

- Helicase is involved in breaking of hydrogen bonds between the two DNA strands
- DNA ligase catalyses phosphodiester bond

**(c)**

- The enzyme required for elongation in DNA replication is DNA polymerase while the enzyme involved in translation is peptidyl transferase
- The bonds catalysed between subunits of monomers in DNA replication is phosphodiester bond while the bonds catalyzed for translation is peptide bonds
- The monomers used for DNA replication is deoxyribonucleotides while the monomers for translation is amino acids

**Question 4****(a) (i) Transformation****(a) (ii)**

- It is a negative control
- to show that bacteria with no ampicillin resistance gene will not grow / multiply

**(a) (iii)**

- bacteria strain is not competent

**(b) (i)**

- The concentration of tryptophan synthesising enzymes decreases while that of lactose utilisation enzymes increases
- The concentration of lactose utilisation enzymes plateau at a higher level than tryptophan synthesizing enzymes

**(b) (ii)**

- Tryptophan binds to trp repressor proteins and activates it
- Trp repressor protein binds to the operator region on the trp operon and turns off the operon.
- Existing tryptophan enzymes are degraded by enzymes

**(c)**

- Permease. ;
- So that the lactose, when present, can be transported into the cells to induce the lac operon and turn on the transcription of more lactose utilisation enzymes. ;

**Question 5****a(i)**

Mitosis

**a(ii)**

- 1M correct plot
- 1M joining the 4 correct dots with a straight line

**(b) (i)**

- The parental stem cell divides to give 2 different cells. One remains as a stem cell while the other differentiate into a specialized cell.

•

**(b) (ii)**

- Hematopoietic stem cell is undifferentiated while its specialised cells are differentiated to have a specific function / structure
- Stem cell can divide and renew itself indefinitely / without limit but red blood cells cannot divide

**c(i)**

missense mutation

**c(ii)**

- 1 nucleotide change from A to T.
- Changes the codon from GAA / GAG, coding for Glu, to GTA / GTG, coding for Val.

**c(iii)**

7

**Question 6****(a)(i)**

- the genes C, D and F are methylated which results in long-term silencing of genes
- Methylated DNA recruits histone deacetylase which removes acetyl group from lysine and arginine residues on histone tail, restores positive charge
- Ionic bonds form between positively-charged histone tails and negatively charged DNA
- Methylation of DNA also changes the conformation of the DNA such that transcription factors and RNA polymerase cannot recognise and bind to access promoter of gene

**(a)(ii)**

- Methylation of gene D for neuron and  $\beta$ -cell but not skin cell allows only the skin cell to produce melanin
- Since all somatic cells contain the same set of genes
- There is differential methylation / gene expression for each specific cell type to carry out its specific function

**(a)(iii)**

- Gene A is an essential gene that is required for normal functioning of all cell types
- For example, gene A codes for RNA polymerase / aminoacyl tRNA synthetase

**(b)**

- The daughter molecule consist of one parental strand with methylated cytosine and one daughter strand.
- Parental strand with methylated cytosine of CG serves as a signal for DMT to methylate the cytosine of CG on the daughter strand

**Question 7****(a)**

$X^O X^O M^L M^L$     $X^O X^O M^L M$     $X^O X^O M M$

$X^O Y M^L M^L$     $X^O Y M^L M$     $X^O Y M M$

1M for every 3 correct

**(b)(i)**

$$\chi^2 = \frac{(17-18)^2}{18} + \frac{(7-6)^2}{6} + \frac{(14-18)^2}{18} + \frac{(10-6)^2}{6}$$

$$\approx 3.78$$

**(b)(ii)**

$$\chi^2 = \frac{(17-16)^2}{16} + \frac{(7-8)^2}{8} + \frac{(14-16)^2}{16} + \frac{(10-8)^2}{8}$$

$$\approx 0.94$$

**c**

- Mary is correct. Although both calculated  $\chi^2$  values are smaller than the critical value, Mary has the smaller calculated  $\chi^2$  value.
- There is a higher probability, that any a difference between the observed and expected number is due to chance, there is a higher probability that the difference is not significant.

**Question 8****(a) (i)**

- Dehydrogenase reduces  $\text{NAD}^+$  to  $\text{NADH}$
- when isocitrate is converted to  $\alpha$ -ketoglutarate / succinyl-CoA; OR when malate is converted to oxaloacetate.
- $\text{NADH}$  carries the electron and proton to electron transport chain for ATP synthesis via oxidative phosphorylation.

OR

- Dehydrogenase reduces  $\text{FAD}^{2+}$  to  $\text{FADH}_2$
- when succinate is converted to fumarate.
- $\text{FADH}_2$  carries the electron and proton to electron transport chain for ATP synthesis via oxidative phosphorylation.

**(a)(ii)**

- Depth B
- Mean dehydrogenase activity was lower, ranging from 1.5-2.5AU, while that of depth A was higher, ranging from approximately 3.2-5.5AU.
- At a greater depth, oxygen concentration is lower. Hence, rate of oxidative phosphorylation is lower.
- Regeneration of  $\text{NAD}^+$  /  $\text{FAD}^{2+}$  is slower hence there is lesser substrates for effective collision.

**(b)**

- lactic acid fermentation
- Pyruvate converted to lactate,  $\text{NADH}$  oxidised to  $\text{NAD}$ .

**(c)(i)**

Carbon dioxide concentration

**(c)(ii)**

- Carbon dioxide concentration is higher near factories
- More  $\text{CO}_2$  for fixation with Ribulose Bisphosphate

**Question 9**

- Variations in population due to random mutation resulting in different alleles;
- primary radiation phase, different niches in the lake with different selection pressure;
- fish with at selective advantage survive and reproduce viable offspring, passing on advantageous genes/alleles to the next generation;
- accumulation of many genetic changes over a long period of time to evolve into different species;
- geographical isolation/ accept hundreds of km apart thus no gene flow between different populations;

4 max



**Question 10****(a)**

- airborne transmission / contact via respiratory droplets.
- when one who is infected with *M. tuberculosis* sneeze / cough and another person inhales the respiratory droplets.
- Only people suffering secondary/reactivated tuberculosis

2max

**(b)(i)**

- The monocytes and dendritic cells are able to present the antigens of *M. tuberculosis* / the bacteria to the CD4 T cells in lymph nodes
- Activated T cells will elicit the adaptive immune response against the bacteria.

**(b)(ii)**

- Cytokines released by infected epithelial cells and macrophages
- Attracts T cells and B cells to the lung parenchyma. ;
- These white blood cells tightly appress the macrophages which are infected by

**Question 11****(a)(i)**

- increasing energy usage which requires the combustion of fossil fuels to generate electricity
- Deforestation results in the removal of forests which are the carbon sink. Burning the forest releases the stored organic carbon back into atmosphere as CO<sub>2</sub>
- Increasing consumption of meat means that carbon dioxide is released indirectly by the agriculture industry rather than the consumers into the atmosphere.

**(a)(ii)**

- atmospheric carbon dioxide increases from 355 to 390ppm, the ocean surface CO<sub>2</sub> concentration for both Hawaii and Bermuda increases from 325 to 360µatm and 335 to 360µatm
- ocean surface CO<sub>2</sub> concentration for Bermuda increases from 335 to 360µatm, pH decreases from 8.105 to 8.078. ocean surface CO<sub>2</sub> concentration for Hawaii increases from 325 to 360µatm, pH decreases from 8.115 to 8.078
- Ocean surface CO<sub>2</sub> dissolves into the ocean water and forms a weak acid which decreases the pH

**(b)**

- increase of surface ocean carbon dioxide from 375ppm to 450-500ppm causes an increase of 2°C water temperature while increase of more than 500ppm causes an increase of 3°C in water temperature
- photosynthesis process in the zooxanthellae is disrupted at higher temperature, and it produces an excess of products that become toxic to itself
- The metabolism of the coral polyp is damaged and expels the zooxanthellae, leaving the coral skeleton 'bleached'
- When there's an increase of 3°C in water temperature, the corals will eventually die from starvation and massive death